

CSE 4203: Discrete Mathematics, Quiz 1 (June 2, 2026)

1. Consider the statement: 'A necessary condition for a graph to be bipartite is that it does not contain an odd cycle.' Let P represent 'The graph is bipartite' and Q represent 'The graph contains an odd cycle.' Write the propositional logic expression that correctly translates this statement? 3 [CO1]

Ans:

2. Write a statement using the predicate and quantifier says that if a number is positive and a second number is greater than the first number, then the second number is also positive? Assume the first and second numbers are x and y , respectively. 3 [CO1]

Ans:

3. What is the correct contrapositive of the English statement: 'If P is true, then Q is true whenever R is false'? 4 [CO1]

Ans:

4. Determine whether $(P \rightarrow Q) \rightarrow (Q \rightarrow P)$ is a tautology using logical equivalences laws of propositional logic. 5 [CO1]

Ans: